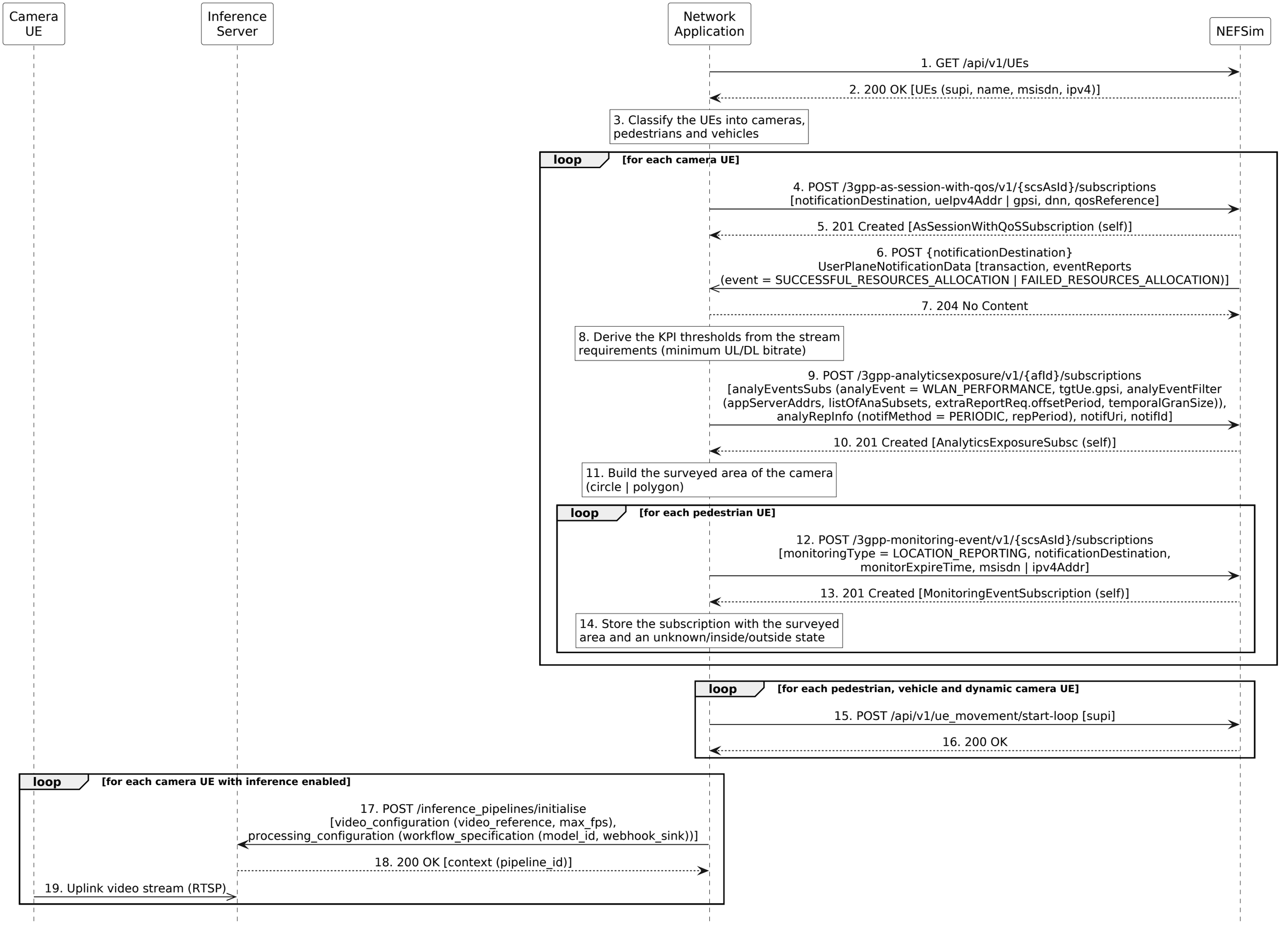
Camera UE
Inference Server
Network Application
NEFSim
1. GET /api/v1/UEs
2. 200 OK [UEs (supi, name, msisdn, ipv4)]
3. Classify the UEs into cameras, pedestrians and vehicles
loop
[for each camera UE]
4. POST /3gpp-as-session-with-qos/v1/{scsAsId}/subscriptions [notificationDestination, ueIpv4Addr | gpsi, dnn, qosReference]
5. 201 Created [AsSessionWithQoSSubscription (self)]
6. POST {notificationDestination} UserPlaneNotificationData [transaction, eventReports (event = SUCCESSFUL_RESOURCES_ALLOCATION | FAILED_RESOURCES_ALLOCATION)]
7. 204 No Content
8. Derive the KPI thresholds from the stream requirements (minimum UL/DL bitrate)
9. POST /3gpp-analyticsexposure/v1/{afId}/subscriptions [analyEventsSubs (analyEvent = WLAN_PERFORMANCE, tgtUe.gpsi, analyEventFilter (appServerAddrs, listOfAnaSubsets, extraReportReq.offsetPeriod, temporalGranSize)), analyRepInfo (notifMethod = PERIODIC, repPeriod), notifUri, notifId]
10. 201 Created [AnalyticsExposureSubsc (self)]
11. Build the surveyed area of the camera (circle | polygon)
loop
[for each pedestrian UE]
12. POST /3gpp-monitoring-event/v1/{scsAsId}/subscriptions [monitoringType = LOCATION_REPORTING, notificationDestination, monitorExpireTime, msisdn | ipv4Addr]
13. 201 Created [MonitoringEventSubscription (self)]
14. Store the subscription with the surveyed area and an unknown/inside/outside state
loop
[for each pedestrian, vehicle and dynamic camera UE]
15. POST /api/v1/ue_movement/start-loop [supi]
16. 200 OK
loop
[for each camera UE with inference enabled]
17. POST /inference_pipelines/initialise [video_configuration (video_reference, max_fps), processing_configuration (workflow_specification (model_id, webhook_sink))]
18. 200 OK [context (pipeline_id)]
19. Uplink video stream (RTSP)